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Magnetism of the Keweenawan age Chengwatana lava flows, northwest Wisconsin

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Abstract. The Chengwatana Volcanics are subaerial plateau basalts which appear to be stratigraphically above the North Shore Volcanic Group of the Lake Superior Region. Sixty five oriented cores were collected at six sites near St. Croix Falls, Wisconsin. Magnetic cleaning of the samples was primarily step-wise alternating field (A.F.) demagnetization to 100 mT. Thermal demagnetization and hysteresis property measurements indicate the dominant magnetic mineralogy is multidomain magnetite. However, a stable remanence is present which we attribute to grains with single domain behavior. The mean remanent direction is D= 287.5°, I= 45.8°, α-95 = 8.0°, with a mean pole at 186.1°E., 30.9°N. This result is consistent with the 1.1 Ga. apparent pole for the North American craton. Two sites mapped as stratigraphically equivalent show distinct reversed directions. Both A.F. and thermal demagnetization isolate the same reversed direction which is interpreted as being primary. These particular flows record a short reversal in an otherwise normal sequence. This strengthens the evidence for more than one reversal in the Middle Keweenawan, and may be the same short reversal noted at Mamainse Point.

Introduction

The Keweenawan Rift Zone dated at about 1.1 Ga. represents one of the largest rift structures in the world, with perhaps the largest volume of basalt flows at a single location and time (Green, 1977). Initial interest in the rift was centered on the copper prospects associated with the secondary mineralization. More recent interest is related to understanding the structure of the rift and developing a detailed stratigraphy of the flow sequences (e.g. Chandler et al., 1990, Paces and Miller,1993).

Our original objective was to use paleomagnetic directions to better constrain the stratigraphic position and structural relationships of the Chengwatana volcanics, since Keweenawan paleopole positions for the craton are well known and a major polarity transition from reverse to normal during the middle Keweenawan has been documented by Palmer (1970), and Halls and Pesonen (1982). Feeney (1990) partially resolved this issue by showing the dominant magnetic polarity of the area was normal, with pole positions consistent with Middle Keweenawan basalts such as the North Shore Volcanics (Books, 1972).

Feeney (1990) also noted several locations with reversed directions. In this study, we report detailed paleomagnetic results of this reversal sequence near St. Croix Falls in Polk County, WI. Magnetic component analysis confirms an interval of reversed directions near the top of the Chegwatana basalt sequence, and attests to the possibility of additional reversals in the Keweenawan as suggested by Palmer (1970), Robertson (1973) and Symons et al. (1994).

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Geology and Previous Paleomagnetism of the Volcanics

The Chengwatana volcanics in Northwest Wisconsin are Keweenawan age volcanic rocks outcropping southeast of the Douglas Fault, and are considered stratigraphically above the North Shore Volcanics (Morey and Mudrey, 1972; Green 1977). The Chengwatana flows dip gently to the southwest (Paces and Miller,1993; Allen and Chandler, 1993), and may be the youngest surface exposure of the Keweenawan volcanics in the Lake Superior region.

The Chengwatana volcanic rocks in Polk Co. are predominately subaerial plateau basalts (Cordua 1989; Cavaleri 1987) that form a series of discontinuous low-lying ridges striking east northeast, over an area of about 10 square kilometers (Figure 1). Cordua (1989) has identified at least eight flow units with a total thickness of 2000 m. Individual flows are between 1 and 30 meters thick. Cavaleri (1987) classified these rocks as tholeiitic basalts, ranging from olivine to quartz tholeiites, and suggests that these basalts represent two distinct eruptive cycles. Geochemical work by Wirth et al. (1995) and Wirth et al. (1997) indicate these lava flows are similar in composition to other middle Keweenawan basalts. Keweenawan rocks generally show some degree of induced thermal alteration at temperatures of over 300°C due to burial (Halls & Pesonen 1982). The basalts near St. Croix Falls have also been subjected to thermal alteration perhaps to temperatures as high as 400°C (Cordua, 1989).

Halls and Pesonen (1982) reviewed the paleomagnetic data from Keweenawan rocks, primarily along the Lake Superior shoreline. Over 60 pole positions are reported which fall within a northeast-southwest band across the North Central Pacific, with a mean pole at 40°N, 180°E, (both polarities are represented). A reversed interval is present in the lower to middle part of the Keweenawan sequence which has been used as a stratigraphic boundary. There is also evidence for an asymmetry of about 30 degrees between the normal and reversed polarities. Palmer (1970) and Robertson (1973) present data for a short reversal in the upper long normal interval at Malaise Point which has not been observed elsewhere.

Feeney's (1990) paleomagnetic study of 18 outcrops in Polk County, western Wisconsin, showed the Chengwatana basalts were predominately normally magnetized, with a mean direction of D=275°, I=41.5°, α-95=15.5°, which is consistent with other Keweenawan directions (Halls & Pacinian). Feeney (1990) also documented a number of reversely magnetized lava flows. Insufficient geologic control over a large sampling area to the north of St.Croix Falls rendered a more precise correlation impossible. However, the volcanics south of St. Croix Falls, WI. are better exposed and have been mapped in detail by Cordua (1989). Supplemental ground magnetic data have also enabled us to cross-correlate the volcanic flows in the several low-lying ridges (Kean et al. 1993a, 1993b). We are thus confident in the stratigraphic relationship of the various flows in this study.

Paleomagnetic Measurements

Sixty five oriented cores were collected at six sites. The sampling sites (Figure 1) were concentrated primarily in the part of the sequence

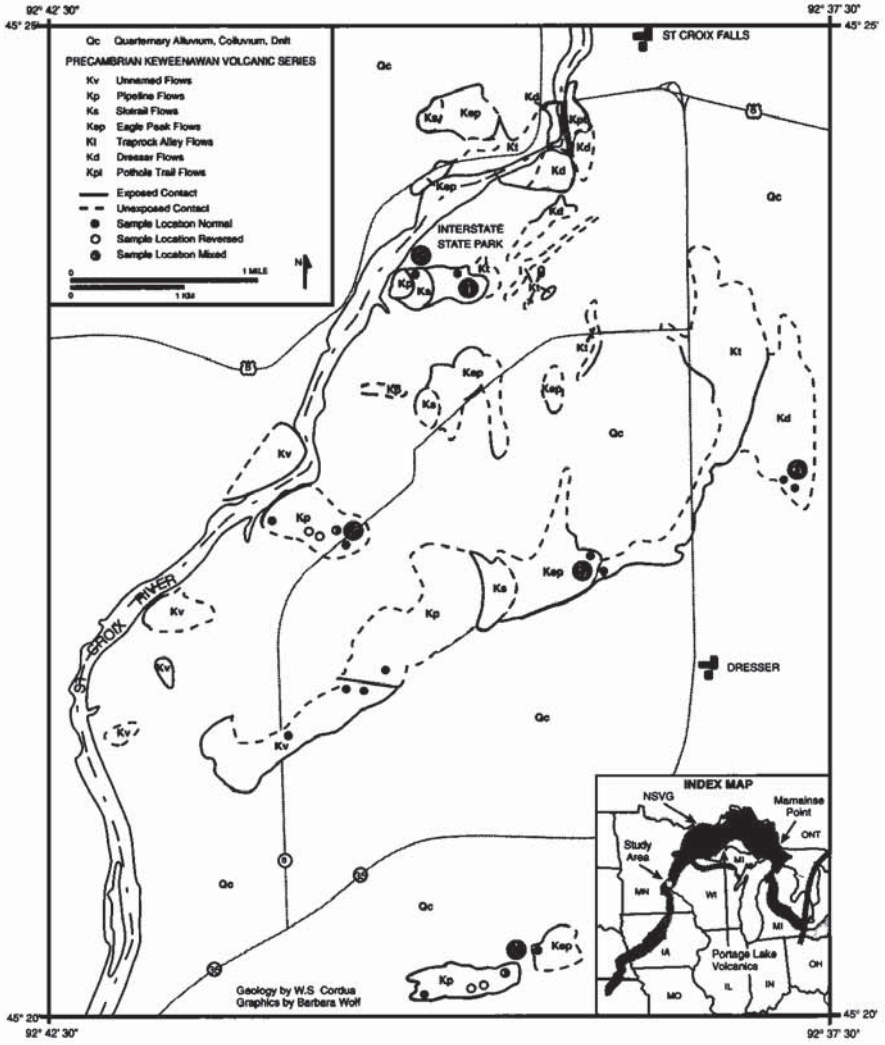


Figure 1. Location map modified from Cordua (1989). The flows dip approximately 10-15 degrees to the west-southwest with the oldest flows to the east. Circles with numbers are sites discussed in this paper. Small closed (open) circles locate normal (reversed) flows.

where Feeney (1990) found reversed directions (sites 1 and 2), and which are considered stratigraphically equivalent. A sun compass was used for orientations in the field to avoid azimuthal errors due to the strong remanent magnetization of the rocks. Most samples were demagnetized using AF fields to 100 mT; a companion set of 19 specimens were thermally demagnetized to 620-700°C. Directional analysis of the magnetic data was carried out using the principal component method described by Kirschvink (1980). Hysteresis properties were determined for 7 samples, and 12 were examined microscopically to determine the degree of chemical alteration, and abundance and nature of magnetic minerals.

Results

Most of the samples have high intensities of natural remanent magnetism (NRM) (10-20 A/m) and volume magnetic susceptibility (4×10^{-2} SI). The magnetic properties differ somewhat between flows, with the upper flows toward the west (sites 1-4) having NRM values and susceptibilities at least an order of magnitude higher than those to the east (sites 5-6). The samples generally carry a viscous component that is removed by AF demagnetization to 15 mT (Figure 2) revealing a characteristic remanent magnetization (ChRM) that decays linearly to the origin with increasing demagnetization. Median destructive fields (MDF) range between 25 and 35 mT. Typical response to thermal demagnetization is illustrated in Figure 3a. The thermal demagnetization curve for specimen SC96-1b (site 2) shows a wide range of unblocking temperatures and a Curie point at 580°C, which is indicative of magnetite as the principal magnetic carrier. More importantly AF and thermal demagnetization of companion specimens from the same core isolate the same ChRM (Figures 3a and 3b).

Hysteresis parameters were determined for samples from each site. Values of coercivity range from 5 to 9.7 mT, with a minimum Hcr/Hc ratio of 2.74, and Jrs/Js ratios from 0.052 to 0.2; both normal and reverse samples are included. Such values are typical of pseudosingle domain and multidomain magnetite (Wasilewski, 1972; Day et

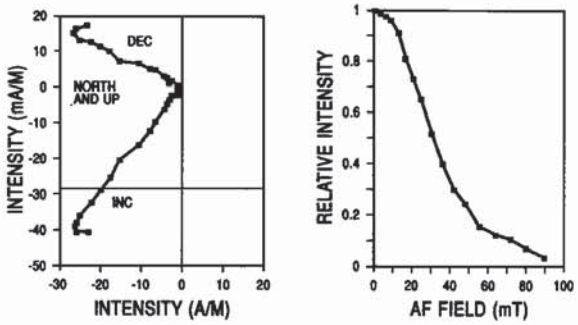


Figure 2. Vector diagram for A.F. demagnetization of normally magnetized specimen 92008b from site 6. Demagnetization is shown from 0 to 100 mT.

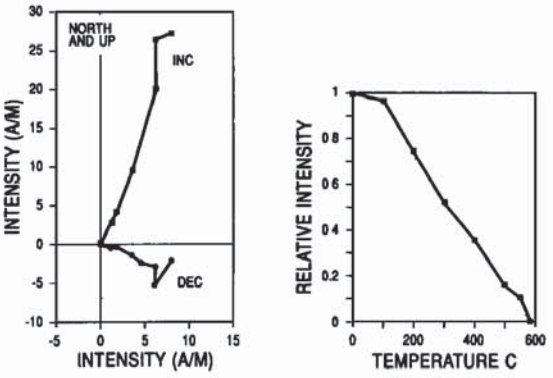


Figure 3a. Vector diagram for thermal demagnetization of reversally magnetized specimen SC96-1b, from site 2. Demagnetization is shown from 20° to 582° C.

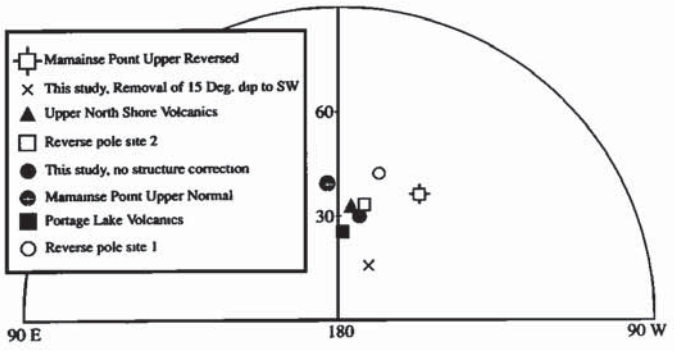


Figure 4. VGP positions for several Keweenawan studies, along with results from this study, both before and after rotation to remove a regional dip of 15 degrees to the southwest. Pole positions for the Portage Lake Volcanics and North Shore Volcanics are averages as reported by Halls and Pesonen (1982). Mamainse Point pole positions are from Palmer (1970).

al.,1977). Although these indicators of magnetic state imply that the samples are dominated by pseudosingle domain and/ or multidomain magnetite, the demagnetization results (Figures 2-3) have led us to conclude that the samples carry a stable remanent magnetization, probably in a small fraction of grains with single domain behavior. Remanence acquisition studies concur that magnetite is the dominant magnetic mineral, although specimens from sites 5 and 6 indicate some hematite as the magnetic carrier.

Principal component analysis (Kirschvink, 1980) reveals two directions that are approximately antiparallel to each other. The most prominent direction, C1, has a northwesterly declination, an inclinations of about 43° (normal magnetization) and is present at all sites. This direction was isolated after the removal of the viscous component (e.g. 92008 Figure 2). A second direction , C2, with a southeasterly declination and approximately -50° inclination (reverse magnetization) is present in samples collected at specific levels near the top of the sequence at sites 1and 2. There are no significant differences either in petrographic characteristics or in the rock magnetic properties between the normal magnetized samples and the reversely magnetized samples at both sites 1 and 2. Wirth et al. (1997) report only minor geochemical differences for the entire sequence of flows. The mean directions (C1 or C2) are regarded in this study as primary directions and are the directions listed in Table 1. Combined data from both components yield a Virtual Geomagnetic Pole (VGP) at 30.9°N, 186°E, which agrees well with published Keweenawan paleopoles for stable North American (Halls and Pacinian, and Figure 4). The results in Table 1 have not been corrected for the regional dip of 10-15 degrees to the west southwest. It was not always possible to measure the local dips of the flows because the bedrock surfaces have been extensively eroded during the Pleistocene glaciation. Instead, the average VGP position is plotted in Figure 5 both before and after a 15° rotation to remove the regional dip. It is evident that the rotation makes little change in the interpretation.

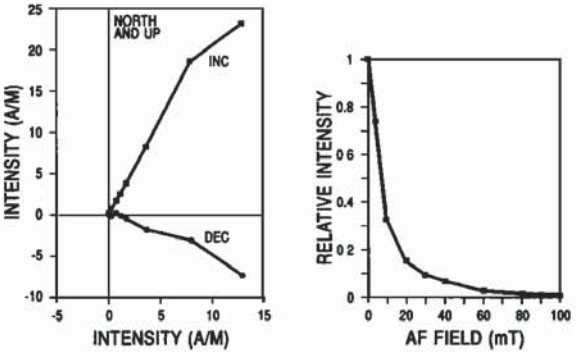


Figure 3b. Vector diagram for A.F. demagnetization of reversally magnetized specimen SC96-1a from site 2. Demagnetization is shown from 20 to 100mT.

Petrographic studies of samples from each site also show differences which are consistent with the differences in susceptibility and intensity. The upper lava flows (sites 1-4) have a high percentage of coarse grained magnetic minerals (25%), which are generally more altered than the older flows (sites 5 and 6). and which appear to have been produced during metamorphism. Sites 5 and 6 have only about 10% magnetite grains which are smaller in size than the younger flows Nonetheless, all these samples contain a ChRM which is probably associated with a small fraction of single domain like grains.

Alpha-95 values for individual sites are in the range of 10-20 degrees which is larger than other studies referenced in Figure 5 and Halls and Pesonen (1982). We attribute this to overlapping coercivity spectra between the original magnetic minerals and the altered minerals. Sites with reversed directions have higher Alpha-95 values (23-29 degrees). This is also seen in thermally demagnetized samples. A total of 19 specimens from sites 1-4 were thermally demagnetized. Five were reversally magnetized (sites 1 and 2) and produced a combined mean direction of D=111° , I = -52° and α-95= 35°. The larger scatter associated with the reversally magnetized samples is most likely due to the long term influence of the dominant normal polarity of the region. However, the metamorphism appears not to have altered the original magnetic directions of any of the flows significantly, since either normal or reversed directions consistent with Keweenawan directions are evident after demagnetization to 20 mT and no samples have NRM directions close to the present day field direction for the study area. However, 4 specimens (sites 1 and 2) showed mixed polarity with A.F. demagnetization. The polarity change was from reversed to normal, in 3 of the specimens and occurred at about 45mT.

There is no strong evidence from our data for asymmetry in the Earth's magnetic field between the normal and reversed polarities as suggested by Halls and Pesonen (1982) for middle Kewenawan rocks. However, we say this with caution, because the alpha-95 cone of confidence for the reversed samples is large enough to overlap with the directions given by Halls and Pesonen (1982). Lastly, our results are consistent with the recent results of Symons et al. (1994) for alkali syenite-carbonitite complexes in the Kapuskasing structural zone in Canada dated at 1097 Ma years, which show more than one reversal in a normal polarity apparent polar wander path (APWP) of similar age. This may in fact be equivalent to the Mamainse Point upper reversal.

Conclusions

A study of six sites in the Keweenawan age Chengwatana lava flow near St. Croix Falls in Polk Co. Wisconsin has provided new information on the reversal record for the middle Keweenawan time.

Table 1. Paleomagnetic data for sites in this study.

						VGP		SITE	
	N	Declination	Inclination	K	Alpha-95	Latitude	Longitude	Latitude	Longitude
Site 1 (N)	10	288.9	55.0	49.5	6.9	N36.7	E193.4	45.33N	92.65W
(R)	7	114.8	-57.8	5.2	29.2	N41	E193.4		
Site 2 (N)	7	274.0	51.1	13	17	N24.6	E198.4	45.37N	92.67W
(R)	8	106.7	-47.5	6.7	23.1	N31.2	E187.9		
Site 3	5	280.0	33.2	14	20.8	N19.9	E183	45.39N	92.66W
Site 4	8	285.1	31	16.3	14.1	N22	E178.9	45.39N	92.66W
Site 5	7	306	45.5	11.2	18.8	N43	E173	45.36N	92.64W
Site 6	8	287.9	42.6	25.4	11.2	N29.6	E183.6	45.37N	92.63W
average normal	6	286.8	43.4	43.9	10.2	N29.7	E184.9		
average N and R	8	287.5	45.8	48.5	8.04	N30.9	E186.1		

The flows exhibit two primary directions which are almost antipodal yielding a combined paleopole of 186.1°E , 30.9° N, which is similar to published middle Keweenawan directions. If a regional dip of 15 degrees to the southwest is removed the pole moves to 188.0° E, 17.5° N.

The major results from this study are first, the discovery of reversely magnetized samples within a normal polarity sequence. The reversal occurs over a stratigraphic distance of tens of meters and in outcrops considered to be stratigraphically equivalent but separated by approximately 2 km. Second, the normal and reversed directions do not show the 30° asymmetry suggested by earlier studies in the Keweenawan basalts. Third, we confirm the presence of more than one reversal in the middle Keweenawan and suggest it is the same short reversal noted at Mamainse Point by Palmer (1970).

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